

Jorge A. Martínez-Ortiz

martjor@msu.edu

Research interests: Machine Learning and Agent-Based Models for Data-Driven Simulation of Complex Systems.

Education

▼ **PhD: Computational Mathematics, Science and Engineering** 08.2023-05.2028 *(expected)*
Michigan State University East Lansing, MI

▼ **BS: Computational Physics** 08.2021-05.2023
Baylor University Waco, TX

Employment

▼ **Graduate Research Assistant** 08.2023-present
Department of Computational Mathematics, Science and Engineering,
Michigan State University East Lansing, MI

- Address: 428 South Shaw Lane 1501 Engineering Building, East Lansing, MI 48824
- Supervisor: Michael S. Murillo

▼ **Graduate Research Intern** 05.2024-08.2024
A-1 Group: Information Systems & Modeling, Los Alamos National
Laboratory Los Alamos, NM

- Address: Bikini Atoll Rd, Los Alamos, NM, United States, 87545
- Supervisor: Katy M. Martinez

▼ **Undergraduate Tutor** 08.2021-05.2023
Center for Academic Success and Engagement, Baylor University Waco, TX

- Address: 1410 S 3rd St, Waco, TX 76798
- Supervisor: Melanie Briscoe

▼ **Undergraduate Research Assistant** 05.2021-05.2023
Center for Astrophysics, Space Physics and Engineering Research, Baylor
University Waco, TX

- Address: 100 Research Pkwy, Waco, TX 76704
- Supervisor: Lorin S. Matthews

Research

▼ **Los Alamos National Laboratory - PI(s): Jim H. Cooley and Katy M. Martinez** 05.2024-present
Graduate Research Intern Los Alamos, NM

- Formulated and tested a family of mathematical models of asymmetric conflict to explicitly account for the psychology and adaptation of agents
- Developed and implemented Python analysis tools that leverage unsupervised machine learning approaches to facilitate model calibration and verification
- Designed efficient modeling workflows to speed up the process of formulating, testing and deploying a model

▼ **Michigan State University - PI(s): Michael S. Murillo** 08.2023-present
Graduate Research Assistant East Lansing, MI

- Developed novel structure-preserving optimization approaches to the reduction of large network datasets.
- Collaborated on the design of efficient AI and data-driven workflows for fast and reliable high-throughput computing.

▼ **Baylor University - PI(s): Lorin S. Matthews and Truell W. Hyde** 08.2021-05.2023
Undergraduate Research Assistant Waco, TX

- Designed MATLAB-C++ hybrid software to enable high-throughput simulation and in-situ visualization of dusty plasmas
- Created and maintained image and video processing workflows for the analysis of experimental dusty plasma data

Publications

1. Watson, Zachary, Parker Adamson, **Jorge Martinez-Ortiz**, Katrina Vermillion, Calvin Carmichael, Samuel Garcia-Rodriguez, Lorin Matthews, Truell Hyde, and Bryant Wyatt. "N-Body Simulation of Spinning Particle Pairs in a Complex Plasma Crystal." *Fundamental Plasma Physics* 13 (March 1, 2025): 100082. <https://doi.org/10.1016/j.fpp.2024.100082>.
2. Carmichael, Calvin, **Jorge Martinez-Ortiz**, Parker Adamson, Lorin Matthews, and Truell Hyde. "Energy Amplification in Plasma Crystals Due to Multiple Torsions." *Physics of Plasmas* 31, no. 12 (December 4, 2024): 123702. <https://doi.org/10.1063/5.0241576>.
3. Carmichael, Calvin, **Jorge Martinez-Ortiz**, Parker Adamson, Lorin Matthews, and Truell Hyde. "Rotating Particle Pair Produces Hot Complex Plasma Crystals." *Physical Review E* 110, no. 2 (August 22, 2024): 025205. <https://doi.org/10.1103/PhysRevE.110.025205>.

Conferences and Presentations

- ▼ **Dimensionality Reduction of Large Network Datasets using Parallel Tempering** 03.2025
SIAM CSE 2025
Poster
- ▼ **Miniaturizing Large Network Datasets for Simulation** 08.2024
2024 AGEF Student Success Conference
Poster
- ▼ **SaLSa: Strategic and Logistic Sandbox for the Modeling of Conflict** 07.2023
Los Alamos National Laboratory 2023 Student Symposium
Poster
- ▼ **Study of mode-coupling instabilities in a plasma crystal using n-body simulations** 08.2022
American Physical Society, Division of Plasma Physics 64th Annual Meeting
Poster
- ▼ **GPU-accelerated n-body simulation of spinning particle pairs in complex plasma crystals** 03.2022
Mathematical Association of America 101st Annual Meeting
Presentation
- ▼ **Dynamic modeling of spinning particle pairs in a complex plasma using NVIDIA GPUs** 03.2022
Nvidia GPU Technology Conference (GTC)
Presentation
- ▼ **Digital representation of fine-grained rim analogs using image processing** 11.2021
American Physical Society, Division of Plasma Physics 63rd Annual Meeting
Presentation
- ▼ **Digital representation of fine-grained rim analogs using image processing** 10.2021
American Astronomical Society, Division for Planetary Sciences 53rd Annual Meeting
Presentation

Awards and Nominations

- ▼ **3rd Best Graduate Student Poster Presentation** 10.2024
2024 AGEF Student Success Conference
- ▼ **Postdoc/Graduate Division Poster Presentation People's Choice Award** 10.2024
2024 AGEF Student Success Conference

Teaching

- ▼ **Physics and Mathematics Undergraduate Tutor** 08.2021-05.2023
Center for Academic Success and Engagement
Baylor University
 - Conducted weekly study sessions for introductory undergraduate physics and mathematics courses
 - Supervised the creation of handouts and study resources provided by the tutoring services to undergraduates
 - Provided one-on-one tutoring to students enrolled in introductory physics and mathematics courses

Coursework

- Michigan State University
 - > GEO 869 Agent-Based Modeling
 - > CMSE 890 Selected Topics: Applied Machine Learning
 - > STT 881 Theory of Probability I
 - > ECE 855 Non-Cooperative Game Theory
 - > CMSE 890 Selected Topics: Reproducible Computational Workflows
 - > CMSE 820 Mathematical Foundations of Data Science
 - > CMSE 822 Parallel Computing
 - > CMSE 821 Numerical Methods for Differential Equations
 - > CMSE 823 Numerical Linear Algebra
- Baylor University
 - > MTH 4328 Numerical Linear Algebra
 - > PHY 4360 Computer Models in Physics
 - > PHY 4340 Statistical and Thermal Physics

Languages

- Spanish: Native
- English: Full Professional Proficiency
- German: Conversational